

武器系统动力学建模与仿真

系统运动微分方程及其数值求解

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系统运动微分方程及其数值求解

- ① 系统动能与广义质量矩阵
- ② 系统动力学微分代数方程
- ③ 系统动力学方程数值求解

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系统动能

The system kinetic energy is defined as

$$T = \sum_{k=1}^N \frac{1}{2} m_k \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k,$$

where

$$\dot{\mathbf{r}}_k = \frac{d}{dt} \mathbf{r}_k(\mathbf{q}(t)) = \sum_{i=1}^n \frac{\partial \mathbf{r}_k}{\partial q_i} \frac{dq_i}{dt} = \sum_{i=1}^n \frac{\partial \mathbf{r}_k}{\partial q_i} \dot{q}_i =: \mathbf{r}_{k,q} \dot{\mathbf{q}}.$$

Then there is

$$T = \sum_{k=1}^N \frac{1}{2} m_k (\mathbf{r}_{k,q} \dot{\mathbf{q}})^T (\mathbf{r}_{k,q} \dot{\mathbf{q}}) = \frac{1}{2} \dot{\mathbf{q}}^T \left(\sum_{k=1}^N m_k \mathbf{r}_{k,q}^T \mathbf{r}_{k,q} \right) \dot{\mathbf{q}}.$$

系统广义质量矩阵

The system kinetic energy could be formulated as

$$T = \frac{1}{2} \dot{\mathbf{q}}^T \left(\sum_{k=1}^N m_k \mathbf{r}_{k,q}^T \mathbf{r}_{k,q} \right) \dot{\mathbf{q}} =: \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}}$$

where the square symmetric matrix $\mathbf{M}(\mathbf{q})$ is called the generalized inertial matrix of the system.

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$$\text{拉格朗日方程: } \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}} \right)^T - \left(\frac{\partial T}{\partial q} \right)^T + c_q^T \lambda = Q$$

The system kinetic energy could be formulated as

$$T = \frac{1}{2} \dot{q}^T M(q) \dot{q},$$

where $M = M^T$. It follows that

$$\begin{aligned} \frac{\partial T}{\partial \dot{q}} &= \dot{q}^T M, \\ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}} \right) &= \ddot{q}^T M + \dot{q}^T \dot{M}. \end{aligned}$$

Substitution into the Lagrange's equation leads to

$$M\ddot{q} + \dot{M}\dot{q} - T_q^T + c_q^T \lambda = Q.$$

拉格朗日方程

The Lagrange's equation in terms of the generalized accelerations

$$M\ddot{\mathbf{q}} + \dot{M}\dot{\mathbf{q}} - T_q^T + \mathbf{c}_q^T \boldsymbol{\lambda} = \mathbf{Q},$$

is often represented as

$$M\ddot{\mathbf{q}} + \mathbf{c}_q^T \boldsymbol{\lambda} = \mathbf{h},$$

where

$$\mathbf{h} = \mathbf{Q} - \dot{M}\dot{\mathbf{q}} + T_q^T.$$

Keep in mind that

$$\mathbf{M}(\mathbf{q}(t)), \quad \mathbf{c}_q(\mathbf{q}(t), t), \quad \mathbf{h}(\mathbf{q}(t), \dot{\mathbf{q}}(t), t).$$

系统动力学微分代数方程

The dynamics of the system is governed by the combination of the Lagrange's equation and the constraint equation of the system, namely

$$\begin{cases} M\ddot{\mathbf{q}} + \mathbf{c}_q^T \boldsymbol{\lambda} = \mathbf{h}, \\ \mathbf{c}(\mathbf{q}, t) = \mathbf{0}, \end{cases}$$

which is a set of differential-algebraic equations (DAE), whose definite solution conditions are usually specified by the corresponding initial values of the generalized coordinates and velocities of the system at the initial moment of time t_0 , denoted as

$$\begin{cases} \mathbf{q}(t)|_{t=t_0} = \mathbf{q}_0, \\ \dot{\mathbf{q}}(t)|_{t=t_0} = \dot{\mathbf{q}}_0. \end{cases}$$

系统运动微分方程及其数值求解

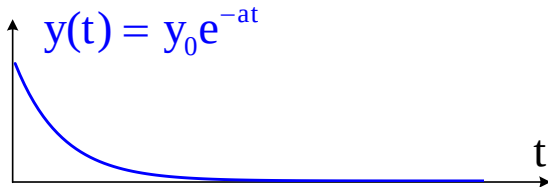
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常微分方程数值求解的计算精度与稳定性

Differential equation governing the water flow of a tank reads as

$$\begin{cases} \dot{y} = -ay, & a > 0, \\ y(t)|_{t=0} = y_0, & y_0 > 0. \end{cases}$$

For this linear differential equation, its analytical solution reads as



which shows that the stored water will run out as time goes by.

常微分方程数值求解的计算精度与稳定性

Differential equation governing the water flow of a tank

$$\dot{y} = -ay, \quad a > 0.$$

When solved numerically, denoting discrete time and states as $t_k = k\Delta t$ and $\hat{y}_k = \hat{y}(t_k)$, respectively, where $k = 0, 1, 2, \dots$, the Taylor expansion reads as

$$\hat{y}_{k+1} = \hat{y}_k + \frac{d\hat{y}_k}{dt}\Delta t + \frac{1}{2!}\frac{d^2\hat{y}_k}{dt^2}\Delta t^2 + \frac{1}{3!}\frac{d^3\hat{y}_k}{dt^3}\Delta t^3 + \frac{1}{4!}\frac{d^4\hat{y}_k}{dt^4}\Delta t^4 + O(\Delta t^4).$$

If keep only the first order (**Computational accuracy is First order**), there is

$$\hat{y}_{k+1} = \hat{y}_k + \left[\alpha \frac{d\hat{y}_k}{dt} + (1 - \alpha) \frac{d\hat{y}_{k+1}}{dt} \right] \Delta t \quad \alpha \in [0, 1].$$

常微分方程数值求解的计算精度与稳定性

Differential equation:

$$\dot{y} = -ay, \quad a > 0.$$

First order Taylor expansion:

$$\hat{y}_{k+1} = \hat{y}_k + \left[\alpha \frac{d\hat{y}_k}{dt} + (1 - \alpha) \frac{d\hat{y}_{k+1}}{dt} \right] \Delta t \quad \alpha \in [0, 1].$$

If let $\alpha = 1$ (Euler Forward, explicit form), then there is

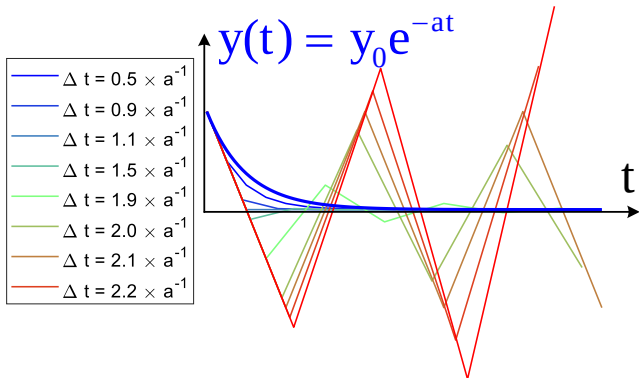
$$\begin{aligned} \hat{y}_{k+1} &= \hat{y}_k + \frac{d\hat{y}_k}{dt} \Delta t \\ &= \hat{y}_k - a\hat{y}_k \Delta t \\ &= (1 - a\Delta t)\hat{y}_k. \end{aligned}$$

常微分方程数值求解的计算精度与稳定性

The explicit form numerical solution

$$\hat{y}_{k+1} = \hat{y}_k - a\hat{y}_k\Delta t \Rightarrow \hat{y}_{k+1} = (1 - a\Delta t)\hat{y}_k$$

exhibits Conditionally Stability.



常微分方程数值求解的计算精度与稳定性

Differential equation:

$$\dot{y} = -ay, \quad a > 0.$$

First order Taylor expansion:

$$\hat{y}_{k+1} = \hat{y}_k + \left[\alpha \frac{d\hat{y}_k}{dt} + (1 - \alpha) \frac{d\hat{y}_{k+1}}{dt} \right] \Delta t \quad \alpha \in [0, 1].$$

If let $\alpha = 0$ (Euler Backward, implicit form), then there is

$$\hat{y}_{k+1} = \hat{y}_k + \frac{d\hat{y}_{k+1}}{dt} \Delta t = \hat{y}_k - a\hat{y}_{k+1} \Delta t,$$

which may be rearranged as

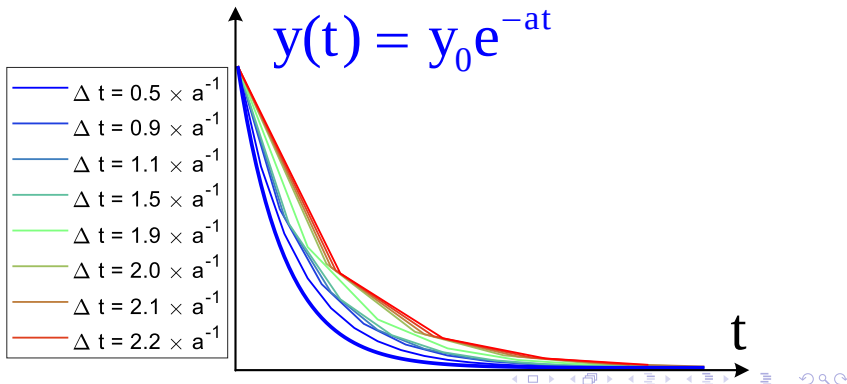
$$\hat{y}_{k+1} = \hat{y}_k / (1 + a\Delta t).$$

常微分方程数值求解的计算精度与稳定性

The implicit form numerical solution

$$\hat{y}_{k+1} = \hat{y}_k - a\hat{y}_{k+1}\Delta t \Rightarrow \hat{y}_{k+1} = \hat{y}_k / (1 + a\Delta t)$$

exhibits Unconditionally Stability.



将高阶常微分方程转化为一阶常微分方程组

Differential equation governing the motion of a pendulum under the effect of gravity may be denoted as

$$ml^2\ddot{\theta} = -mgl \sin \theta,$$

which is a second-order nonlinear ordinary differential equation, whose initial conditions may be denoted as

$$\begin{cases} \theta(t)|_{t=t_0} = \theta_0, \\ \dot{\theta}(t)|_{t=t_0} = \dot{\theta}_0. \end{cases}$$

Typically, its numerical solution may be calculated by reformulating the second-order ordinary differential equation as a set of first-order nonlinear ordinary differential equations.

将高阶常微分方程转化为一阶常微分方程组

Motion equation of a single pendulum

$$ml^2\ddot{\theta} = -mgl \sin \theta.$$

Augmented as system of first order differential equations

$$\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} y_2 \\ -g \sin y_1/l \end{bmatrix}, \quad \begin{cases} y_1 \triangleq \theta, \\ y_2 \triangleq \dot{\theta}. \end{cases}$$

Given the initial state of the system (initial position and initial velocity)

$$\begin{cases} y_1(t)|_{t=0} = y_1(0) \\ y_2(t)|_{t=0} = y_2(0) \end{cases},$$

there exists various sophisticated numerical algorithms to be adopted.

一阶常微分方程组初值问题的一般形式

一阶常微分方程组的初值问题的一般形式可记为

$$\begin{cases} \mathbf{y}' = \mathbf{f}(t, \mathbf{y}), t \in [a, b], \\ \mathbf{y}(a) = \mathbf{y}_0. \end{cases}$$

式中

$$\mathbf{y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_m(t) \end{bmatrix} \in R^m, \quad \mathbf{f} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_m \end{bmatrix} \in R^m, \quad \mathbf{y}_0 = \mathbf{y}(t)|_{t=a} \in R^m$$

分别为待求的状态变量列阵 (解向量), 已知的状态变化率函数 (向量函数) 和已知的状态变量列阵初值 (初始向量).

一阶常微分方程组初值问题数值解法 (一)

线性多步法 (Linear multistep method):

$$\mathbf{y}_n = \sum_{i=1}^k \alpha_i \mathbf{y}_{n-i} + h \sum_{i=0}^k \beta_i \mathbf{f}(t_{n-i}, \mathbf{y}_{n-i}),$$

当 $\beta_0 = 0$ 时为显式公式, 当 $\beta_0 \neq 0$ 为隐式公式. 当 $k = 1$ 时退化为单步法.

一阶常微分方程组初值问题数值解法 (二)

预估校正法 (Predictor corrector method):

$$\bar{\mathbf{y}}_n = \sum_{i=1}^k \bar{\alpha}_i \mathbf{y}_{n-i} + h \sum_{i=1}^k \bar{\beta}_i \mathbf{f}(t_{n-i}, \mathbf{y}_{n-i}), \quad (\text{I})$$

$$\mathbf{y}_n = \sum_{i=1}^k \alpha_i \mathbf{y}_{n-i} + h \sum_{i=1}^k \beta_i \mathbf{f}(t_{n-i}, \mathbf{y}_{n-i}) + h\beta_0 \mathbf{f}(t_n, \bar{\mathbf{y}}_n). \quad (\text{II})$$

式 (I) 为显式预估公式, 式 (II) 为隐式校正公式.

一阶常微分方程组初值问题数值解法 (三)

龙格库塔法 (Runge-Kutta method):

$$\mathbf{y}_n = \mathbf{y}_{n-1} + h \sum_{i=1}^s b_i \mathbf{k}_i, \quad (\text{I})$$

$$\mathbf{k}_i = \mathbf{f}(t_{n-1} + c_i h, \mathbf{y}_{n-1} + h \sum_{j=1}^s a_{i,j} \mathbf{k}_j). \quad (\text{II})$$

如果当 $j \geq i$ 时 $a_{i,j} = 0$, 式 (I) 和 (II) 就是显式 RK 方法. 如果当 $j > i$ 时 $a_{i,j} = 0$, 而对象元素 $a_{i,i}$ 不全为 0, 式 (I) 和 (II) 就称为 0 隐式 RK 方法; 若此时 $a_{i,i}$ 均相等, 则称为对角隐式 RK 方法.

一阶常微分方程组初值问题数值解法 (三)

经典四阶定步长龙格库塔法 (Classical four-order fixed-step Runge-Kutta method):

$$\mathbf{y}_n = \mathbf{y}_{n-1} + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + \mathbf{k}_3 + \mathbf{k}_4),$$

$$\mathbf{k}_1 = \mathbf{f}(t_{n-1}, \mathbf{y}_{n-1}),$$

$$\mathbf{k}_2 = \mathbf{f}(t_{n-1} + \frac{h}{2}, \mathbf{y}_{n-1} + \frac{h}{2}\mathbf{k}_1),$$

$$\mathbf{k}_3 = \mathbf{f}(t_{n-1} + \frac{h}{2}, \mathbf{y}_{n-1} + \frac{h}{2}\mathbf{k}_2),$$

$$\mathbf{k}_4 = \mathbf{f}(t_{n-1} + h, \mathbf{y}_{n-1} + h\mathbf{k}_3).$$

微分代数方程初值问题数值求解

Recall the differential-algebraic equations governing the system dynamics

$$\begin{cases} M\ddot{\mathbf{q}} + \mathbf{c}_q^T \boldsymbol{\lambda} = \mathbf{B}, \\ \mathbf{c}(\mathbf{q}, t) = \mathbf{0}, \end{cases}$$

and its initial values specified by

$$\begin{cases} \mathbf{q}(t)|_{t=t_0} = \mathbf{q}_0, \\ \dot{\mathbf{q}}(t)|_{t=t_0} = \dot{\mathbf{q}}_0. \end{cases}$$

Differentiating the position-level constraint equation

$$\mathbf{c}(\mathbf{q}, t) = \mathbf{0},$$

with respect to time leads to velocity-level constraint equation

$$\mathbf{0} = \mathbf{c}_q(\mathbf{q}(t), t)\dot{\mathbf{q}}(t) + \mathbf{c}_t(\mathbf{q}(t), t),$$

where

$$\mathbf{c}_q(\mathbf{q}(t), t) = \begin{bmatrix} \frac{\partial c_1}{\partial q_1} & \frac{\partial c_1}{\partial q_2} & \cdots & \frac{\partial c_1}{\partial q_n} \\ \frac{\partial c_2}{\partial q_1} & \frac{\partial c_2}{\partial q_2} & \cdots & \frac{\partial c_2}{\partial q_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial c_m}{\partial q_1} & \frac{\partial c_m}{\partial q_2} & \cdots & \frac{\partial c_m}{\partial q_n} \end{bmatrix}, \quad \mathbf{c}_t(\mathbf{q}(t), t) = \begin{bmatrix} \frac{\partial c_1}{\partial t} \\ \frac{\partial c_2}{\partial t} \\ \vdots \\ \frac{\partial c_m}{\partial t} \end{bmatrix}.$$

Further differentiating the velocity-level constraint equation

$$\mathbf{0} = \mathbf{c}_q(\mathbf{q}(t), t)\dot{\mathbf{q}}(t) + \mathbf{c}_t(\mathbf{q}(t), t),$$

with respect to time leads to acceleration-level constraint equation

$$\mathbf{0} = \mathbf{c}_q\ddot{\mathbf{q}} + (\mathbf{c}_q\dot{\mathbf{q}})_q\dot{\mathbf{q}} + 2\mathbf{c}_{qt}\dot{\mathbf{q}} + \mathbf{c}_{tt} =: \mathbf{c}_q\ddot{\mathbf{q}} + \boldsymbol{\varepsilon}(\mathbf{q}, \dot{\mathbf{q}}, t),$$

where

$$\mathbf{c}_{qt} = \begin{bmatrix} \frac{\partial^2 c_1}{\partial q_1 \partial t} & \frac{\partial^2 c_1}{\partial q_2 \partial t} & \cdots & \frac{\partial^2 c_1}{\partial q_n \partial t} \\ \frac{\partial^2 c_2}{\partial q_1 \partial t} & \frac{\partial^2 c_2}{\partial q_2 \partial t} & \cdots & \frac{\partial^2 c_2}{\partial q_n \partial t} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 c_m}{\partial q_1 \partial t} & \frac{\partial^2 c_m}{\partial q_2 \partial t} & \cdots & \frac{\partial^2 c_m}{\partial q_n \partial t} \end{bmatrix}, \quad \mathbf{c}_{tt}(\mathbf{q}(t), t) = \begin{bmatrix} \frac{\partial^2 c_1}{\partial t^2} \\ \frac{\partial^2 c_2}{\partial t^2} \\ \vdots \\ \frac{\partial^2 c_m}{\partial t^2} \end{bmatrix}.$$

系统动力学方程

Global dynamics equation for multibody systems

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{c}_q^T \boldsymbol{\lambda} = \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}, t),$$

where

$$\mathbf{M}(\mathbf{q}, t) \in \mathbb{R}^{n \times n}, \mathbf{q}(t) \in \mathbb{R}^n, \mathbf{c}_q(\mathbf{q}, t) \in \mathbb{R}^{m \times n}, \boldsymbol{\lambda} \in \mathbb{R}^m, \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}, t) \in \mathbb{R}^n.$$

Constraint equations:

$$\mathbf{0} = \mathbf{c}(\mathbf{q}(t), t),$$

$$\mathbf{0} = \dot{\mathbf{c}}(\mathbf{q}(t), \dot{\mathbf{q}}(t), t) = \mathbf{c}_q \dot{\mathbf{q}} + \mathbf{c}_t,$$

$$\mathbf{0} = \ddot{\mathbf{c}}(\mathbf{q}(t), \dot{\mathbf{q}}(t), \ddot{\mathbf{q}}(t), t) = \mathbf{c}_q \ddot{\mathbf{q}} + \boldsymbol{\varepsilon}.$$

二阶常微分方程组形式的系统动力学方程

Combining of the system dynamics equation

$$M\ddot{\mathbf{q}} + \mathbf{c}_q^T \boldsymbol{\lambda} = \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}, t),$$

and the acceleration-level constraint equation

$$\mathbf{0} = \mathbf{c}_q \ddot{\mathbf{q}} + \boldsymbol{\varepsilon},$$

leads to

$$\begin{bmatrix} M & \mathbf{c}_q^T \\ \mathbf{c}_q & \mathbf{O} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{h} \\ -\boldsymbol{\varepsilon} \end{bmatrix}.$$

When solved explicitly, the preceding equation may be equivalently denoted as

$$\ddot{\mathbf{q}} = \mathbf{f}_{\text{acc}}(\mathbf{q}, \dot{\mathbf{q}}, t).$$

一阶常微分方程组形式的系统动力学方程

The system of second order differential equations

$$\ddot{\mathbf{q}} = \mathbf{f}_{\text{acc}}(\mathbf{q}, \dot{\mathbf{q}}, t),$$

may be augmented as a system of first order differential equations

$$\begin{bmatrix} \dot{\mathbf{y}}_1 \\ \dot{\mathbf{y}}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{y}_2 \\ \mathbf{f}_{\text{acc}}(\mathbf{y}_1, \mathbf{y}_2, t) \end{bmatrix}. \quad \begin{cases} \mathbf{y}_1 \triangleq \mathbf{q} \\ \mathbf{y}_2 \triangleq \dot{\mathbf{q}} \end{cases}.$$

Numerical solution may cause of constraint violation on Position and Velocity level.

违约与修正

Assume that the constraint equation are slightly violated but has increased greater than specified threshold

$$\mathbf{0} \approx \mathbf{c}(\mathbf{q}(t), t).$$

An deviation of $\mathbf{q}$, denoted as $\Delta\mathbf{q}$, with minimum 2-norm or ∞ -norm, are expected to eliminate the constraint violation on position level, denoted as

$$\mathbf{0} = \mathbf{c}(\mathbf{q}(t) + \Delta\mathbf{q}, t).$$

This may be accomplished by incorporating Newton-Rapson iteration and generalized inverse theory. Analogues procedure applies for evaluating of $\Delta\dot{\mathbf{q}}$, but without iteration, to eliminate the constraint violation on velocity level.